

CSARCH2-G8-GPU-WARS

Title: GPU Wars: Five Turning Points That Changed Computing

GitHub Link: <https://github.com/K-K-R-C/CSARCH2-G8-GPU-WARS/blob/main/README.md>

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Topic Theme

Every frame rendered in a video game, every AI-generated image and every large language model relies on a technology that was originally designed for a much simpler purpose: drawing graphics faster.

The Graphics Processing Unit (GPU) has undergone one of the most significant architectural transformations in computing history. What began as dedicated hardware for 3D graphics evolved into a massive computing platform that powers artificial intelligence, scientific simulations and modern super computers.

This exhibit explores five major turning points that shaped the evolution of the GPU and examines how it became one of the most important technologies in modern computing.

Tech Stack

Frontend Technologies

Technology	Role
Astro 6	Primary framework for exhibit development and integration with the museum website
React (JSX)	Interactive educational components and visualizations
MDX	Combines exhibit content with embedded React components
CSS	Styling, animations and responsive layouts

- **Astro 6**

Astro will serve as the foundation of the virtual exhibit. It provides compatibility with the museum template, while also allowing for interactive React components to be integrated within MDX content.

Planned uses:

1. Exhibit page rendering
2. Component integration
3. Static site generation

- **React (JSX)**

React will be used to build the interactive elements that allow visitors to explore GPU history and architecture.

Planned uses:

1. Interactive Timeline
2. GPU Architecture Explorer
3. GPU Generation Comparison Tool

- **MDX**

MDX will be used to combine written explanations with interactive components in a seamless exhibit format.

It will support:

1. Historical Narratives
2. Technical explanations
3. Embedded interactive visuals

- **CSS**

CSS will be used to create a modern digital museum aesthetic inspired by GPU hardware and computing systems.

Design features include:

1. Dark-themed interface
2. Card-based layouts
3. Smooth transitions and animations
4. Responsive desktop/mobile design

Backend Technologies

Technology	Role
Node.js	Ensures compatibility with modern JavaScript tooling and supports deployment of the exhibit website.
Express	Will only be used if needed for additional functionality in future development.

- **Node.js**

Node.js will be used as the local runtime environment required by Astro for development and build processes.

- **Express**

Possible uses:

1. Storing user interaction data (e.g., GPU comparison results)
2. Logging exploration activity
3. Supporting future expansion of interactive components

I. Exhibit Narrative

1.) Introduction: The Rise of GPU

Today, GPUs power video games, artificial intelligence, and scientific simulations. However, they were originally designed to solve a much narrower problem: rendering graphics efficiently.

This exhibit explores how that specialized hardware evolved into one of the most important computing platforms in modern history.

2.) Before Dedicated Graphics Hardware

Before GPUs existed, all graphics processing was handled by the CPU.

Topics covered include:

- CPU-based rendering
- Software rasterization
- Performance bottlenecks
- Increasing demand for 3D graphics

Key Idea: As graphical demands increased, general-purpose CPUs became insufficient for real-time rendering workloads.

3.) Five Turning Points in GPU History

This section is the core of the exhibit and is presented as an interactive timeline.

a. Turning Point #1 - 1996

3dfx Voodoo Graphics

- First widely successful consumer 3D accelerator

b. Turning Point #2 - 1999

GeForce 256

- Introduction of the “GPU” concept with hardware transform and lighting

c. Turning Point #3 - 2007

CUDA

- GPUs become general-purpose parallel computing devices

Turning Point #4 - 2018

RTX 20 Series

- Introduction of ray tracing and AI-assisted rendering hardware

Turning Point #5 - 2022-Present

AI Supercomputing Era

- GPU’s become core infrastructure for AI training and inference

4.) Evolution of GPU Architecture

This section explores the major architectural developments that transformed GPUs from specialized graphics accelerators into powerful platforms for scientific computing, real-time rendering, and artificial intelligence.

Architectural Progression

```
Fixed-Function Pipeline
↓
Transform and Lighting Hardware
↓
Programmable Shaders
↓
CUDA (General-Purpose Computing)
↓
RT Cores (Ray Tracing)
↓
Tensor Cores (AI Acceleration)
```

Key Takeaway

- Hardware acceleration
- Parallel processing
- Programmable hardware design
- Specialized compute units
- Throughput-oriented architecture

5.) GPU Competition (“GPU Wars”)

GPU evolution was strongly shaped by competition between hardware manufacturers.

Topics covered include:

- 3dfx Vs. NVIDIA (Early 3D graphics era)

- NVIDIA Vs. ATI/AMD (GPU standardization era)
- NVIDIA Vs. AMD Vs. Intel (AI computing era)

Key Idea: Competition between companies accelerated architectural innovation and pushed GPUs into new computing domains.

6.) Conclusion and Legacy

The GPU is now a foundational component of modern computing systems.

Without GPUs:

- Modern 3D graphics would not exist at scale
- Scientific simulations would be significantly slower
- Machine learning would be impractical
- Large language models would not be feasible

Final Statement:

The GPU did not just improve graphics performance, but it also fundamentally changed the nature of computing.

II. Computer Architecture Concepts

This exhibit directly connects GPU history to core CSARCH principles.

a. Hardware Acceleration

Specialized hardware improves performance for specific workloads.

b. Parallel Processing

GPUs execute thousands of operations simultaneously.

c. Programmable Architectures

Transition from fixed-function pipelines to flexible compute systems.

d. Specialized Processing Units

Introduction of RT Cores and Tensor Cores.

e. Throughput-Oriented Design

Optimization for massive data processing rather than single-task speed.

III. Visitor Experience and Interactive Features

1.) Interactive Timeline (Main Component)

Visitors explore GPU history through a clickable timeline of major milestones.

Features:

- Historical explanations

- Visual diagrams
- Architectural breakdowns
- Key innovations per era

2.) GPU Architecture Explorer

Visitors can select a GPU generation and explore its architecture.

Each selection shows:

- Primary purpose
- Key architectural features
- Supported workloads
- Major innovations

Possible comparisons: | # | Comparison | |---|-----| | 1 | Voodoo Graphics (1996) | | 2 | GeForce 256 (1999) | | 3 | CUDA-era GPUs | | 4 | RTX GPUs | | 5 | AI accelerators |

3.) GPU Generation Comparison Tool (possible future component)

A side-by-side comparison tool where users select two GPU generations to compare.

Example Comparisons: | # | Comparison | |---|-----| | 1 | GeForce 256 vs. RTX GPU | | 2 | GTX Era vs. H100 | | 3 | Gaming GPU vs. AI GPU |

Displayed data:

- Year of release
- Intended purpose
- Architecture type
- Memory system
- Specialized hardware
- Typical workloads

Educational Goal:

Helps users visually understand how GPU architecture evolved over time in terms of complexity, specialization and computing capability.

IV. Visual Design Direction And Layout

Theme: Modern Technology Museum / GPU Architecture Exhibit

Key Design Features

- Dark semiconductor-inspired aesthetic
- Card-based information panels
- Interactive timeline animations

- Circuit-themed visual accents

Planned Display

V. Planned React Components

Component	Purpose
InteractiveTimeline.jsx	Main timeline of GPU milestones
TimelineEvent.jsx	Individual timeline entries
ArchitectureExplorer.jsx	GPU architecture breakdown viewer
GPUComparisonTool.jsx	Side-by-side GPU generation comparison
GPUInfoCard.jsx	Displays structured GPU details